

Assessing Hierarchies and Part-to-Whole Relationships Transcript

Now, we move to Assessing Hierarchies. As I mentioned earlier, assessing hierarchies is important to understand part to whole relationship. You have a phenomenon which is happening at the aggregate level, but there are some phenomena or subcategories which contribute to the overall phenomenon.

You want to understand how these two are related and what is the role played by each component in the overall scheme of things or the whole part, how the whole is formed upon the parts or how the parts are composed in the whole. We can answer these questions using visuals that deal with part to whole hierarchies. Here we are talking about an interesting case called Circle Packing Diagram.

It is driven by particular softwares which are not readily available in our excel sheets, but are available in advanced tools like Tableau or Power BI or in Python. We have to import these things and do the visualisations. However, we will try to see what is that they are trying to convey and how they can be used in particular cases.

In this case, we are looking at the educational spending done by Gates Foundation on different programmes. So, the size of the circle indicates the share of the spending in the overall spending and also the different types of programmes that they are running. Green circles indicate a particular programme compared to light green and the red circles.

So, this in this way you are not only categorising the spending, but also showing the size of the spending and the overall circle indicates the total amount of spending. And you can readily see the projects that are in the centre occupy higher amount of spending on a particular type of activity. And once you can divide that by year, you can understand how the circle is composed also and the text box can contain the information about that particular unit.

In this way circle packing diagram is showing how smallest of the units are contributing to the whole and how the whole is composed of particular large projects. This is very interesting to show let us say you have 100 business regions in your company, you can use this chart to show the sales contribution by each region over a period of time and see how each region is moving. So, this is an interesting way to see the part to whole relationships.

The another part to whole relationship can be explored using the bubble hierarchy. In this case as done with the circle to part whole relationship, there is a central bubble which represents the whole from which offshoots are drawn and each offshoot is again subdivided into internal categories indicating its overall relationship to the whole. So, we can understand there is a central bubble in this case and it is given to some amount more than 10 categories and each category size representing its share in the whole and the subsequent bubbles from that category indicate the subdivisions in this.

So, this is very interesting to see the use of bubble hierarchy to understand how



individual components are related to the whole part and how whole part is related to the individual components. Again as in the case in the last time if your company has 100 sub regions each has a further subdivisions, you can represent the entire sales revenue generated from them using a bubble hierarchy and understand which regions are important or contributing most to the sales and which regions are smaller ah in comparison to the whole. You can also get insights on which 2 regions are comparable or which 2 projects are comparable and in relation to the whole you can always make such comparisons readily through this chart.

These charts as I mentioned earlier are done through special softwares ah and they need to be customised for specific cases. We are just trying to explain them so, that you are exposed to these elements and you can use them for customising your data visualisation. Thank you.

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